

Summary Data on Acid Cropland and Lime Requirements in Kenya

GAIA

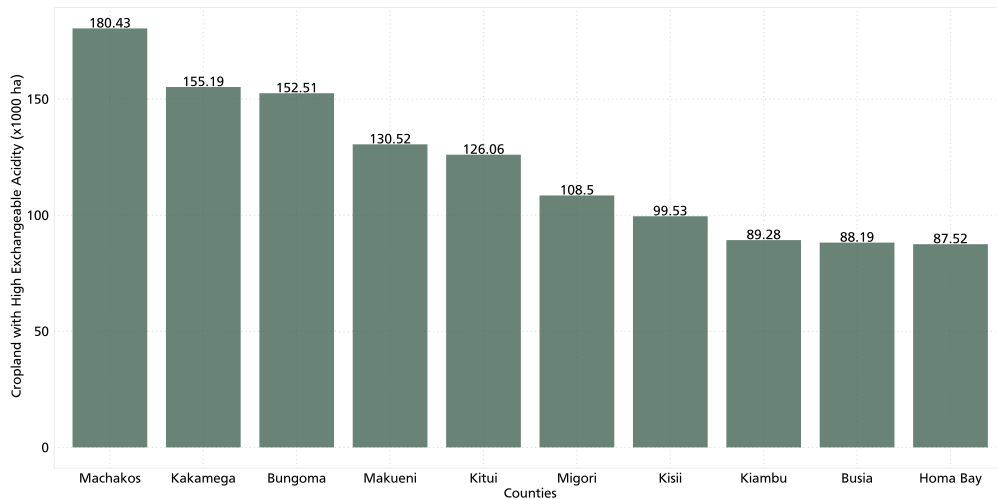
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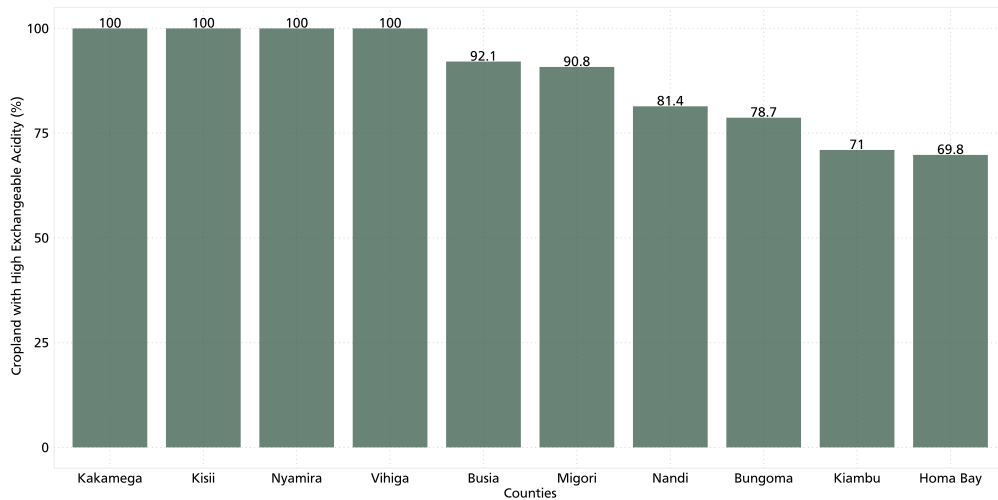
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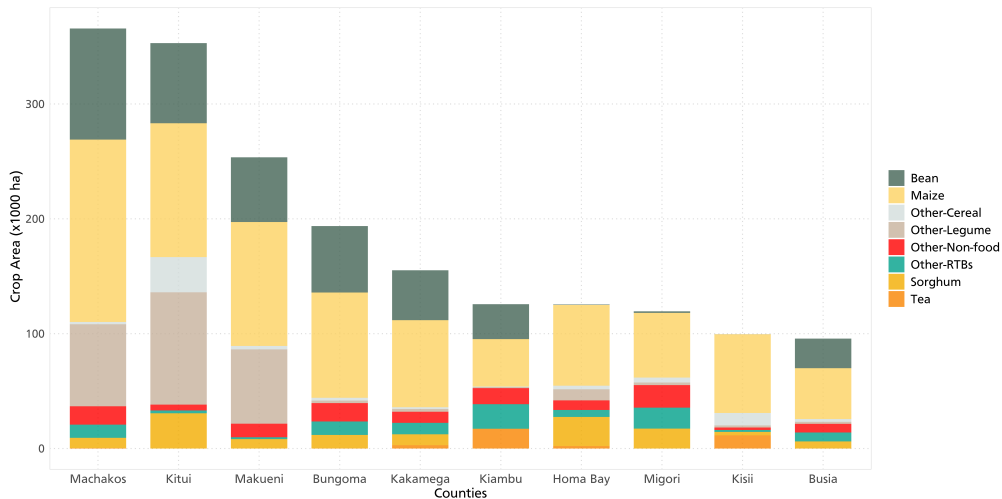
Top 10 Counties in Kenya with the highest acidic cropland



Top 10 Counties in Kenya with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Counties, not segregated by acidic cropland



Kenya-Summary

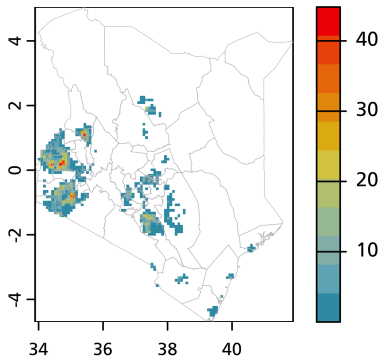
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Counties, not segregated by acidic cropland

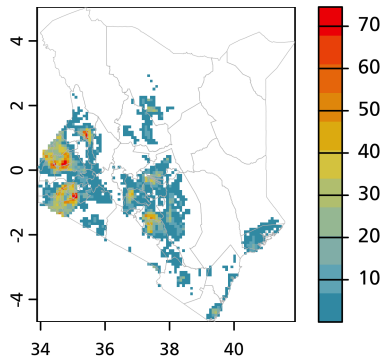
Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum	Tea
Bungoma	57.88	91.63	2.46	2.17	16.02	11.69	11.74	0.11
Busia	25.79	44.29	2.39	1.76	7.52	7.78	6.16	0.03
Homa Bay	0.46	70.27	3.15	9.63	8.31	6.11	25.39	2.11
Kakamega	43.44	75.78	1.50	2.48	9.64	9.96	9.53	2.86
Kiambu	30.46	41.28	0.47	0.92	13.97	21.37	0.09	17.14
Kisii	0.09	68.52	10.79	1.86	2.10	1.70	2.99	11.48
Kitui	69.77	116.53	30.68	97.76	5.08	2.48	30.68	-
Machakos	96.77	158.79	1.99	71.39	15.99	11.50	8.94	0.37
Makueni	56.32	107.91	2.99	64.68	11.84	1.63	8.16	-
Migori	1.41	56.27	4.37	2.25	19.58	18.24	17.37	-

Yield loss due to soil acidity for Maize and Bean in Kenya

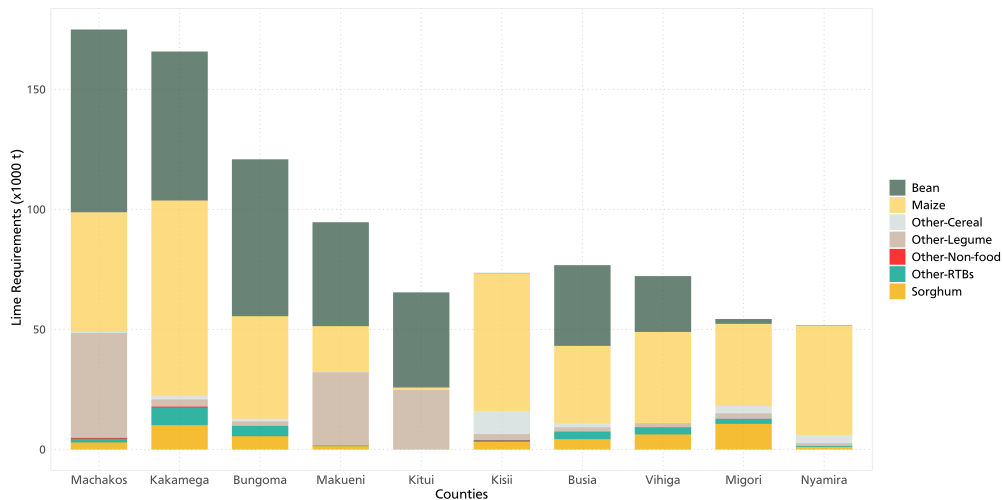
Maize Yield Loss (%)



Bean Yield Loss (%)



Lime requirements (t) for acid soil remediation in Kenya

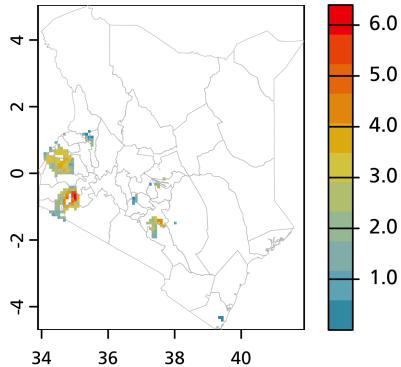


Lime requirements (t) for acid soil remediation in Kenya (table)

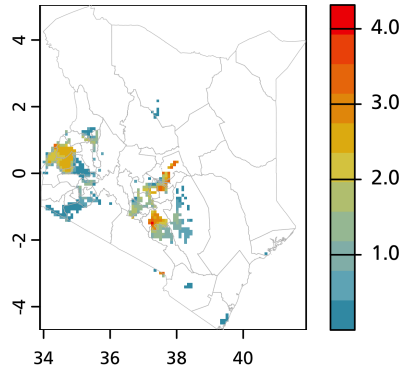
Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-RTBs	Sorghum	Other-Non-food
Bungoma	65.36	42.71	1.12	1.90	4.27	5.55	-
Busia	33.60	32.27	1.67	1.80	3.07	4.35	0.03
Kakamega	62.05	81.26	1.62	3.01	7.37	10.17	0.32
Kisii	0.12	57.36	9.66	2.49	0.62	3.16	0.18
Kitui	39.62	1.00	0.07	24.58	-	0.20	-
Machakos	76.14	49.81	0.54	43.67	1.56	2.97	0.31
Makueni	43.27	19.01	0.21	30.51	0.26	1.41	-
Migori	2.01	34.20	3.12	2.25	2.12	10.69	-
Nyamira	0.19	45.56	3.37	1.15	0.39	1.09	0.01
Vihiga	23.27	37.92	0.11	1.67	3.01	6.28	-

Lime rate for Maize and Bean in Kenya

Maize Lime Rate (t/ha)

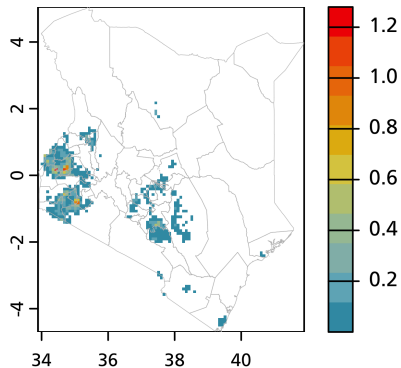


Bean Lime Rate (t/ha)

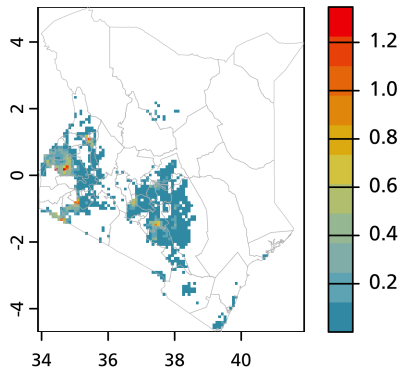


Yield response to lime application in Kenya

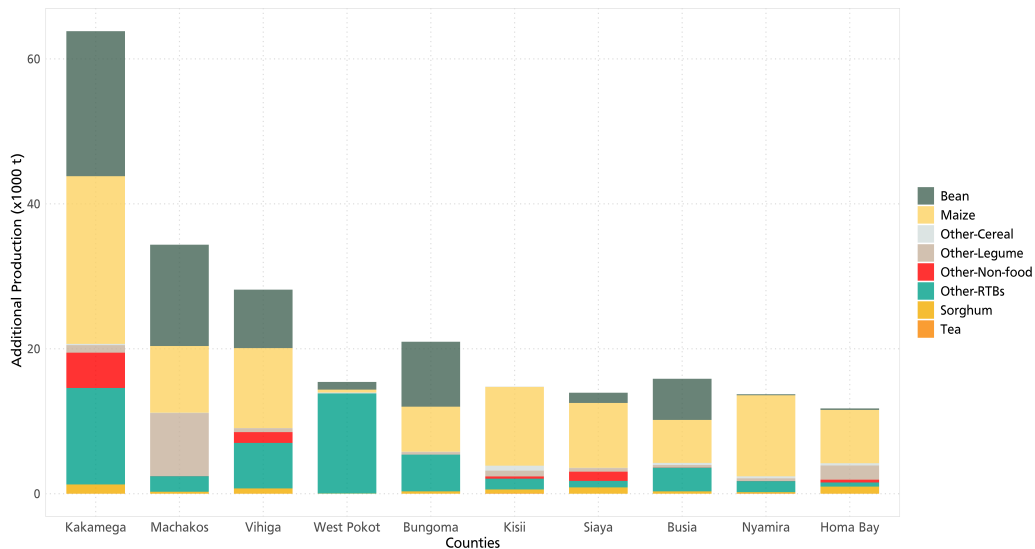
Maize Yield Response (t/ha)



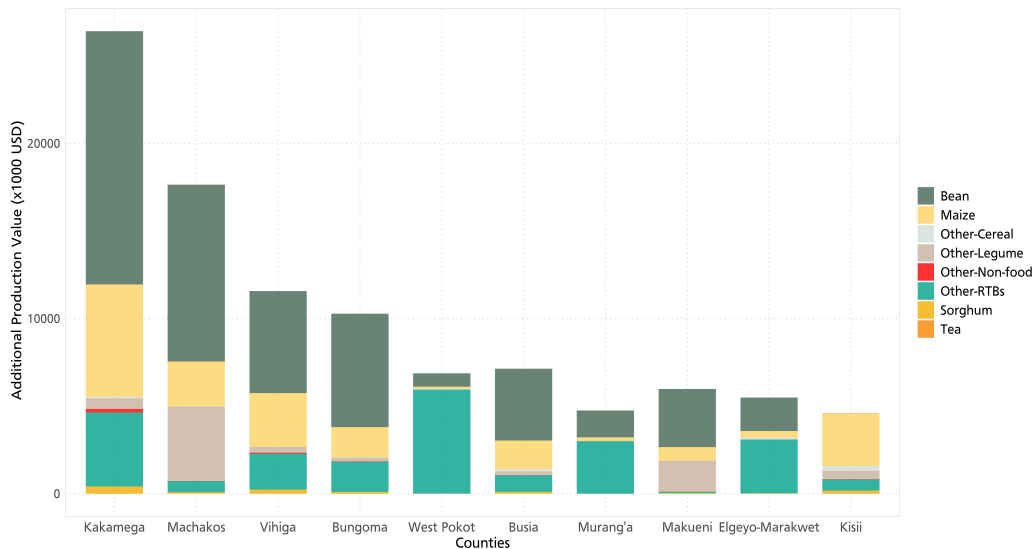
Bean Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Counties in Kenya

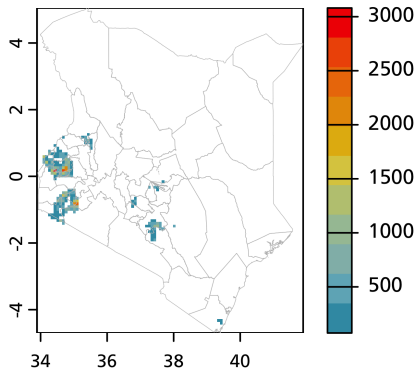


Potential for additional production value (US\$) in top 10 acidic Counties in Kenya

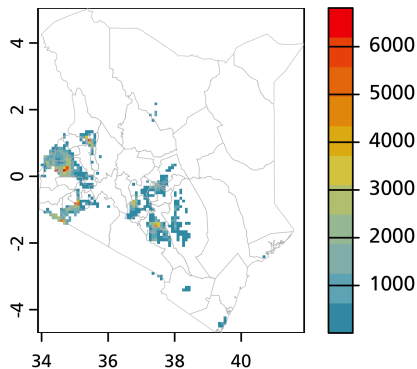


Return on investment for lime application in Kenya

Maize ROI (USD/ha)

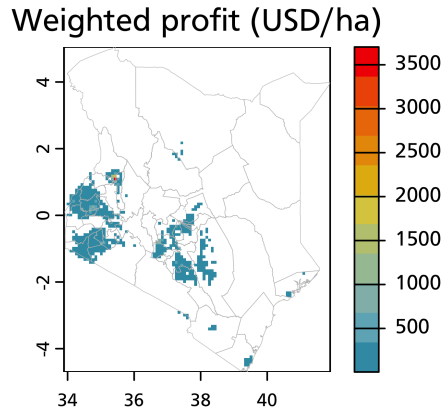
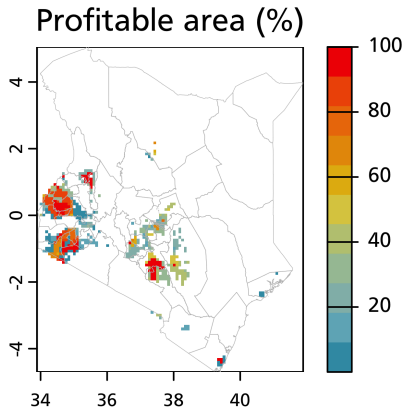


Bean ROI (USD/ha)



- └ Profitability of Lime Application
- └ Profitability of lime application in Kenya

Profitability of lime application in Kenya



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.